

PanUtil User Manual

AudioForge PanUtil v1.0.0

Professional stereo panning and width control with phase correlation metering

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Overview

PanUtil is a precision stereo imaging tool that provides:

- **Stereo panning:** Position sounds in the stereo field (-100% L to +100% R)
- **Width control:** Expand or narrow the stereo image (0% to 200%)
- **Phase correlation meter:** Visualize mono compatibility
- **Constant-power panning:** Maintains perceived loudness when panning
- **Balance mode:** Alternative panning algorithm for stereo sources

When to Use PanUtil

- **Stereo placement:** Position sounds in the left-right stereo field
 - **Stereo width adjustment:** Make tracks wider or narrower
 - **Mono compatibility checking:** Ensure mix translates to mono playback
 - **Mid/side processing:** Control center vs sides independently
 - **Creative effects:** Extreme width or phase manipulation
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Quick Start

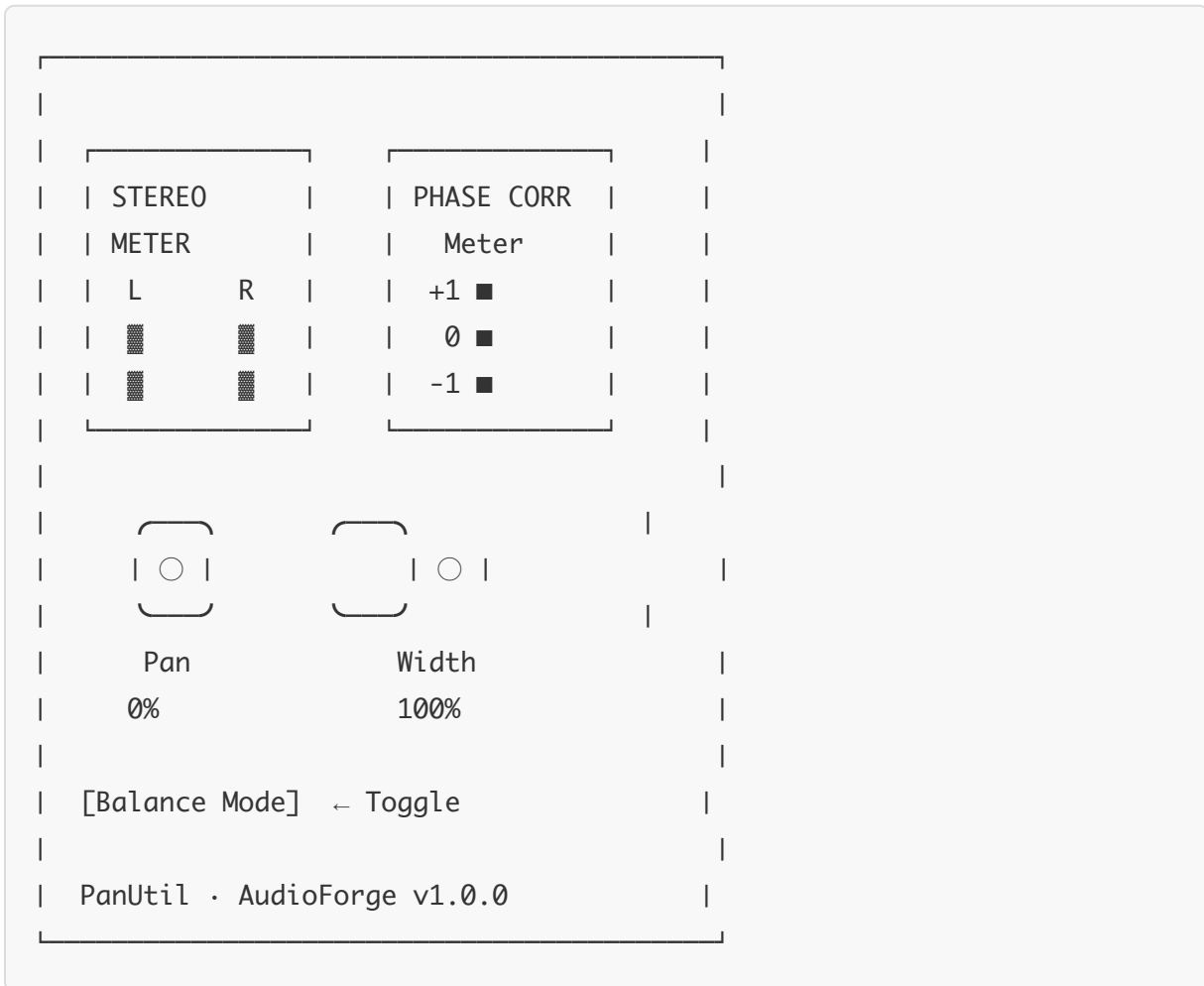
Basic Panning

1. **Load PanUtil** on a mono or stereo track
2. **Adjust Pan knob:**
 - Turn left for left positioning
 - Turn right for right positioning
 - Center (0%) for center placement
3. **Watch the stereo meter** to visualize positioning

Width Control

1. **Load PanUtil** on a stereo track
 2. **Adjust Width knob:**
 - 100% = Original width (no change)
 - <100% = Narrower (more mono)
 - >100% = Wider (more stereo separation)
 3. **Watch phase correlation meter** to avoid phase issues
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User Interface



UI Elements

Pan Knob:

- **Range:** -100% (hard left) to +100% (hard right)
- **Default:** 0% (center)
- **Double-click** to reset to center

Width Knob:

- **Range:** 0% (mono) to 200% (ultra-wide)
- **Default:** 100% (original)
- **Double-click** to reset to 100%

Stereo Meter:

- Shows L/R channel levels in real-time
- Visual feedback for pan position

Phase Correlation Meter:

- **+1** (green): Perfect mono compatibility, in-phase
- **0** (yellow): Some phase difference
- **-1** (red): Phase-reversed, will cancel in mono

Balance Mode Toggle:

- **Off** (default): Pan mode (constant-power)
- **On**: Balance mode (for stereo sources)

Parameters

Pan

Range: -100% to +100%

Default: 0% (center)

Description:

Controls the left-right stereo position of the audio signal.

Pan Mode (default):

- Uses constant-power panning law (-3 dB in center)
- Maintains perceived loudness when panning
- Best for mono sources

Balance Mode:

- Adjusts relative level of L/R channels
- Preserves stereo image of stereo sources
- Best for stereo sources (loops, samples, groups)

Technical Details:

- **-100%**: Full left (L = 1.0, R = 0.0)
 - **0%**: Center (L = 0.707, R = 0.707) [-3 dB each]
 - **+100%**: Full right (L = 0.0, R = 1.0)
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Width

Range: 0% to 200%

Default: 100% (original)

Description:

Controls the stereo width/separation of the signal.

Width Values:

- **0%**: Mono (L and R summed to center)
- **100%**: Original stereo width (no change)
- **150%**: 50% wider than original
- **200%**: Double the original width (maximum)

Technical Implementation:

Uses mid/side (M/S) processing:

- **Mid**: Center/mono content (L+R)
- **Side**: Stereo content (L-R)
- Width control adjusts the Side level

Warning: Values >150% can cause phase issues. Always check phase correlation meter.

Use Cases

1. Stereo Field Arrangement

Goal: Position multiple elements across the stereo field for clarity and separation.

Technique:

- **Center** (0%): Lead vocal, kick, snare, bass
- **Slight left/right** ($\pm 20\text{-}30\%$): Rhythm guitars, keys, secondary vocals
- **Wide** ($\pm 60\text{-}80\%$): Hi-hats, cymbals, pads, effects
- **Hard pan** ($\pm 100\%$): Doubled guitars, stereo effects, ear candy

Why it works: Creates space, prevents masking, makes mix feel wide and spacious.

2. Stereo Width Enhancement

Goal: Make a track sound wider/bigger without losing mono compatibility.

Technique:

1. Start with Width at 100%
2. Gradually increase to 120-140%
3. **Watch phase correlation meter:**
 - Stay above 0 for mono compatibility
 - If it goes below 0, reduce width

When to use:

- Synth pads (makes them lush and spacious)
- Stereo reverbs (adds depth)
- Background vocals (makes them feel "around" the lead)

When NOT to use:

- Bass (keep it mono/centered)
 - Kick/snare (lose power when widened)
 - Lead vocals (can lose focus)
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3. Mono Compatibility Checking

Goal: Ensure your mix sounds good on mono playback (phones, club systems, Bluetooth speakers).

Technique:

1. Adjust width on various tracks
2. **Watch phase correlation meter:**
 - **+0.5 to +1.0:** Safe for mono (good)
 - **0 to +0.5:** Some phase cancellation (check in mono)
 - **<0:** Significant cancellation (fix by reducing width or adjusting pan)

Fix phase issues:

- Reduce width to 80-90%
 - Adjust panning of clashing elements
 - Flip phase on one track if needed
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4. Haas Effect (Stereo Widening)

Goal: Create psychoacoustic width using timing differences.

Technique (requires 2 instances of PanUtil):

1. **Track 1:** Pan 100% left, no width change
2. **Track 2** (duplicate): Pan 100% right, delay by 10-30ms
3. Result: Wide stereo image from mono source

Note: PanUtil alone doesn't add delay - combine with a delay plugin for Haas effect.

5. Mid/Side Processing

Goal: Control center vs sides independently.

Width as M/S control:

- **Increase width:** Boosts sides, makes track spacious
- **Decrease width:** Boosts mid, makes track focused

Creative uses:

- **Vocals:** Reduce width to 80-90% for focused, centered sound
- **Pads:** Increase width to 130-150% for lush, enveloping feel
- **Guitars:** Pan left/right, then increase width for "wall of sound"

Tips & Techniques

1. LCR Panning (Left-Center-Right)

Technique: Only pan to hard left, center, or hard right (no in-between).

Benefits:

- Maximum clarity and separation
- Works well for rock, metal, and dense mixes
- Easy to visualize and manage

How to use PanUtil:

- Center: 0%
 - Hard left: -100%
 - Hard right: +100%
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2. Stereo Widening Without Phase Issues

Rule of thumb: Stay below 150% width for most sources.

Safe widening:

1. Set width to 120-140%
 2. Check phase correlation meter (should stay >0.3)
 3. A/B test in mono (solo track, toggle DAW mono button)
 4. If it sounds thin in mono, reduce width
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3. Use Balance Mode for Stereo Sources

When you have:

- Stereo loops
- Stereo samples
- Stereo recorded instruments (acoustic guitars, pianos)

Why: Balance mode preserves the internal stereo image while adjusting L/R position.

Toggle: Click "Balance Mode" button in UI.

4. Narrow Bass for Focus and Power

Technique:

1. Insert PanUtil on bass track
2. Reduce width to 0-50%
3. Result: Focused, centered bass with more punch

Why: Bass frequencies below ~200 Hz should be mostly mono for:

- Maximum power and impact
 - Mono compatibility
 - Better translation to small speakers
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5. Wide Pad/Synth Trick

Technique:

1. Duplicate synth/pad track
2. **Track 1:** Pan -30%, Width 120%
3. **Track 2:** Pan +30%, Width 120%
4. Result: Massive, surrounding stereo image

Bonus: Detune Track 2 by +5-10 cents for chorus-like effect.

6. Check Phase Correlation Constantly

Best practice: Always keep an eye on the phase correlation meter.

Target values:

- **Master bus:** +0.7 to +1.0 (excellent mono compatibility)
 - **Individual tracks:** +0.3 to +1.0 (acceptable)
 - **Warning zone:** 0 to +0.3 (check in mono)
 - **Problem:** <0 (phase issues, will sound thin in mono)
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Technical Specifications

Audio Processing

- **Bit Depth:** 32-bit floating point
- **Sample Rate:** 44.1 kHz - 192 kHz
- **Latency:** 0 samples
- **Pan Law:** -3 dB constant-power (pan mode)
- **Width Algorithm:** Mid/side processing

Performance

- **CPU Usage:** < 0.2% (negligible)
- **RAM Usage:** < 1 MB

Metering

- **Stereo Meter:** Peak level, L/R independent
- **Phase Correlation:** -1.0 (phase-reversed) to +1.0 (perfect mono)
- Update rate: 50ms
- Smoothing: 200ms

Compatibility

- **Formats:** VST3, AU (macOS), Standalone
- **Platforms:** macOS 10.13+, Windows 10/11, Linux

- **DAWs Tested:** Ableton, Logic, Reaper, FL Studio, Cubase, Bitwig
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Troubleshooting

Mix sounds thin in mono

Cause: Excessive width or phase issues.

Fix:

1. Check phase correlation meter (should be >0.3)
 2. Reduce width on tracks with values $>140\%$
 3. A/B test in mono to identify problematic tracks
 4. Consider using less stereo widening plugins overall
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Pan doesn't seem to work

Cause: Balance mode enabled on mono source, or subsequent plugin is re-centering.

Fix:

1. Disable Balance Mode (use Pan mode for mono sources)
 2. Check if other plugins after PanUtil are affecting pan
 3. Verify track is actually receiving audio (check meters)
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Phase correlation meter always shows +1.0

Cause: Track is mono (both L/R identical).

Fix:

- This is normal for mono tracks
 - Width control will have no effect on pure mono sources
 - If you want stereo from mono, add stereo delay, chorus, or reverb first
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Support

For installation, bug reports, and feature requests, see [SimpleGain Manual - Support section](#).

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